

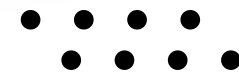
Fox & Hounds Game

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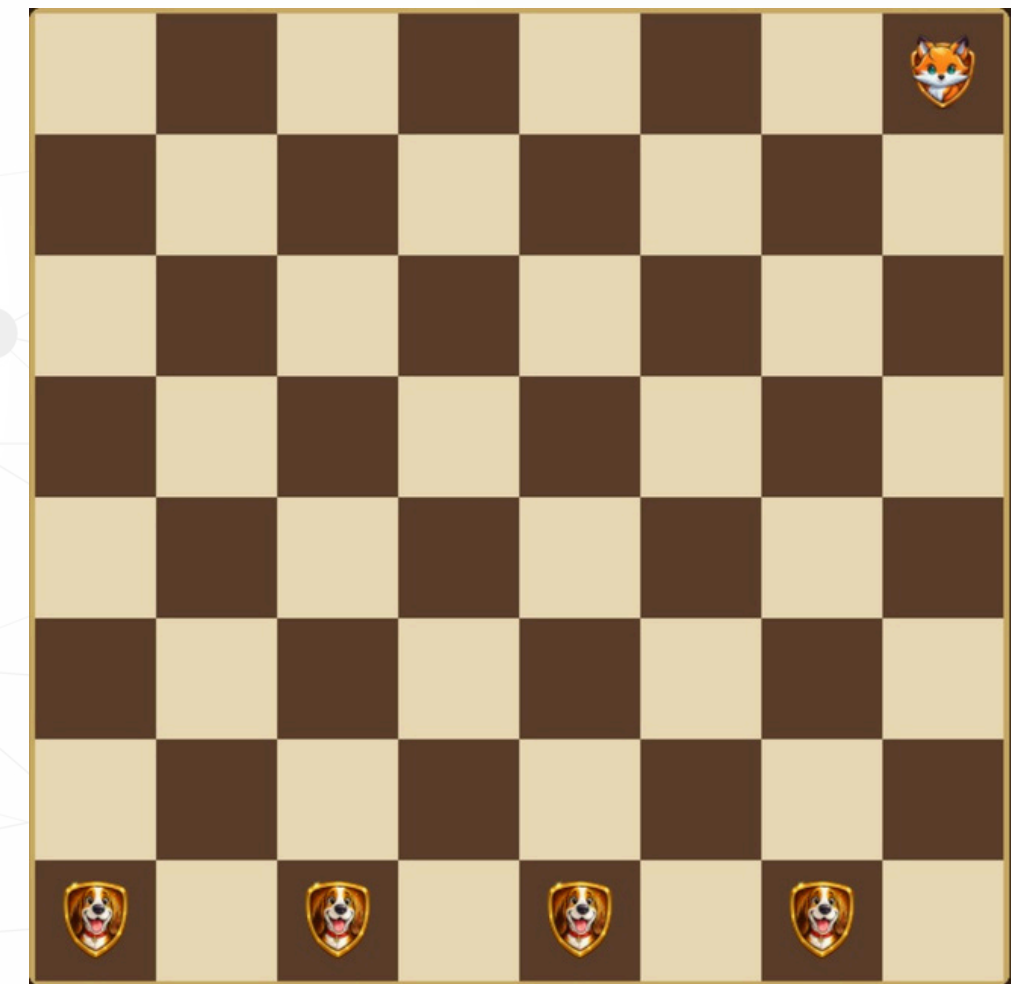
Let's Get Started





How to play

- Fox and Hounds is a strategy game played on an 8×8 checkerboard between one fox and four hounds.
- There are two modes in the game. A player can choose to play either as the fox or as the hounds, while the other player controls the remaining side.
- At the start, the four hounds are placed on one side of the board and the fox on the opposite side. In this version, the game can start on either the dark or light brown squares.
- All pieces move diagonally one square at a time. The hounds can only move forward, while the fox can move forward and backward.
- The fox wins by reaching the opposite side of the board, while the hounds win by trapping the fox so it cannot move



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Project Goal

To build an AI agent that can intelligently play Fox and Hounds using adversarial search.

01

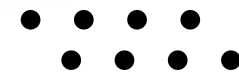
Minimax

02

Alpha-Beta
pruning

03

Heuristic
evaluation
function



State Of The Art VS Our Approach

State-of-the-art AI uses:

- deep search
- advanced heuristics
- learning-based methods

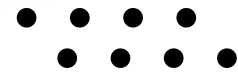
Our implementation:

- Minimax with Alpha-Beta pruning
- heuristic evaluation function

Focus was on balancing:

- Speed
- decision quality

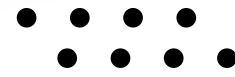




Game Implementation

- Built using Python + Pygame for an interactive experience
- Built an interactive GUI-based board system
- Designed a dynamic menu and control interface
- **Developed:**
 - rule-based legal move generation system
 - turn management logic
 - win condition detection
- Integrated smooth animations for improved user experience





Evaluation Function

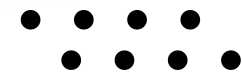
Evaluation Function

- A heuristic function used to evaluate non-terminal states
- Guides the Minimax algorithm at limited search depth

Features used:

- Fox progress toward winning
- Mobility of the fox
- Strategic positioning of hounds
- Distance-based pressure on the fox
- Trap detection for near-winning states





Why we chose this approach

Fox and Hounds is a strategic adversarial game

Suitable for:

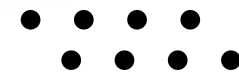
- Minimax algorithm
- Alpha-Beta pruning

Allows exploration of:

- decision-making under limited depth
- heuristic design

Helps understand how AI evaluates game states





Most Interesting Idea

Even without deep search, the AI behaves intelligently just because of the evaluation function.

Even with limited search depth:

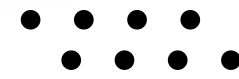
- a good heuristic can guide better decisions

Small changes in evaluation logic:

- significantly change AI behavior

Shows how AI “thinks” about the game





Implementation VS Idea

Idea:

- AI explores all possible moves deeply
- always finds optimal strategy

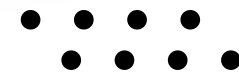
Implementation:

- limited depth due to performance constraints
- relies on heuristic evaluation

Trade-off:

- accuracy vs speed

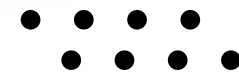




Most Striking Success

- AI made more strategic moves after improving evaluation function
- Hounds were better at trapping the fox
- Fox mobility was effectively reduced
- Alpha-Beta pruning improved performance
- Game became more realistic and competitive

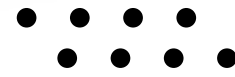




Most Striking Failure

- AI still makes suboptimal decisions at low depth
- Cannot always predict long-term strategies
- Sometimes misses better moves due to heuristic limitations
- Performance decreases with higher depth





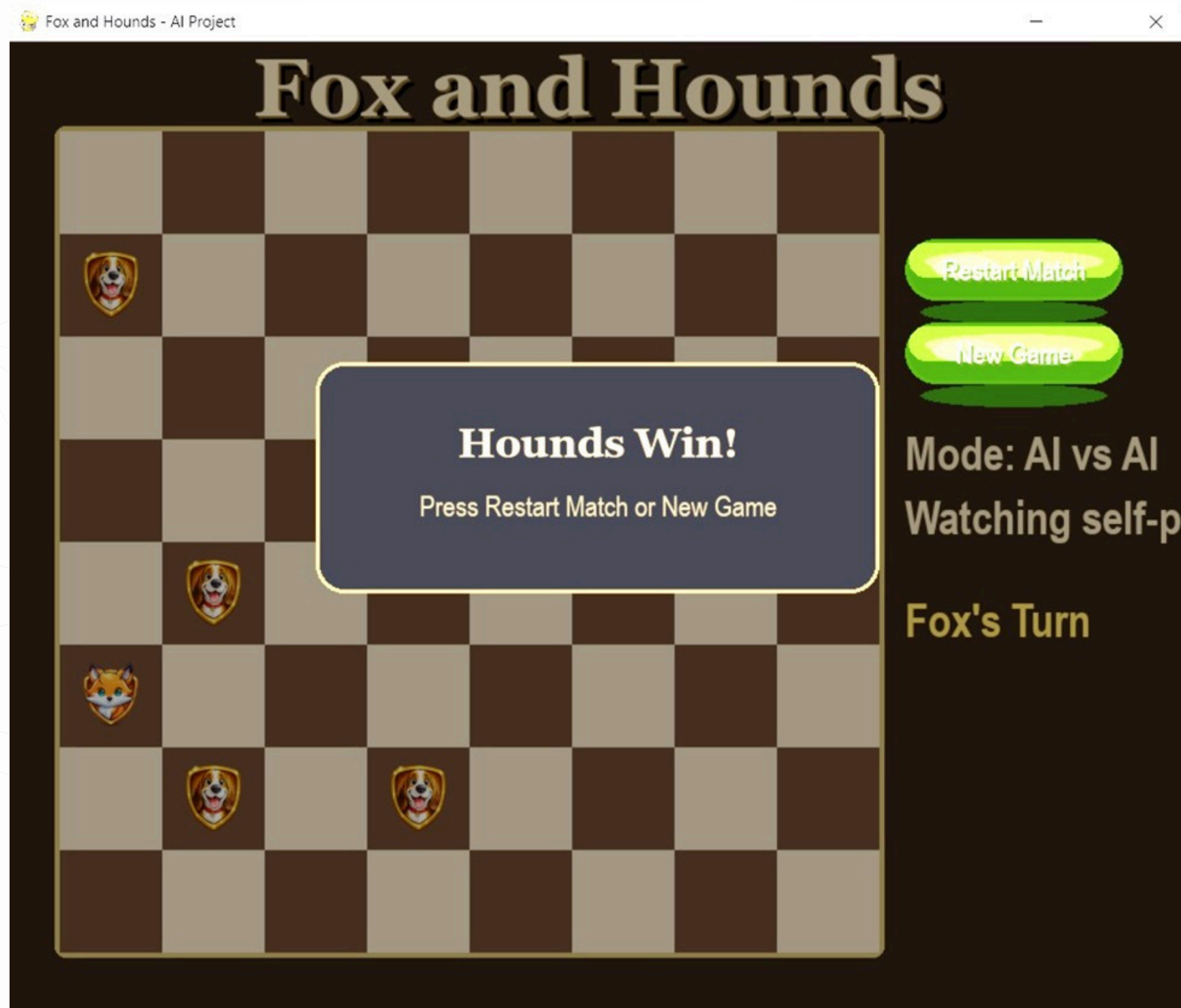
Suggested Improvements

- Enhance the evaluation function by considering:
 - open escape paths for the fox
 - blocking strategies for hounds
- Fine-tune heuristic weights to balance mobility, positioning, and trapping
- Increase minimax depth while maintaining performance using optimizations
- Reduce randomness in move selection to improve consistency
- Implement AI vs AI simulations to evaluate long-term performance
- Explore learning-based approaches (e.g., reinforcement learning) for adaptive AI



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Problems we faced (Before)



Fox was losing every time in the same scenario.

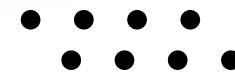


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Problems we faced (After)



Improved the code so they are not ending up in same scenario.



Training First 5 Games

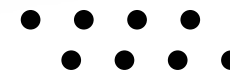


Shows variation and learning behavior

Fox Progress: 10
Hound Dist: 12.0
Hound Ahead: 15.0

Fox Wins: 0
Hound Wins: 5
Moves This Match: 9
Training Left: 6

Recent History:
Hound won | W:10/12.0/15.0



Training 80+ Games



Fox win rate increased from 10% to 45%



Thank You!